Word Embeddings and the Brain

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Abstract

This study extends Pereira et al. (2018)'s neural decoding work through structured analyses comparing embedding types and an open-ended cross-lingual investigation. We compared Word2Vec and GloVe for concept decoding (comparable performance), evaluated contextualized versus static embeddings on sentence decoding (Sentence-BERT achieved 97.77% vs. GloVe's 80.16% on experiment 2), and implemented brain encoder models (BERT predicted 693 vs. GloVe's 61 voxels with $R^2 > 0.1$).

Our primary contribution addresses whether conceptual brain representations transfer across languages. Training decoders on English fMRI data from 180 concepts, we tested cross-lingual transferability to Arabic, Hebrew, Spanish, and French using multilingual BERT. Results revealed successful cross-lingual transfer with mean ranks of 76.79-85.12 (compared to chance=90, and English baseline=80.67), with systematic language family patterns. Romance languages excelled at action concepts, Semitic at abstract psychological concepts, and Germanic at descriptive properties. These findings support constrained universality in semantic brain representations, where basic conceptual patterns transfer cross-culturally but linguistic factors create systematic encoding differences. [1]

Keywords: neural decoding; cross-lingual; fMRI; word embeddings; conceptual representation; multilingual BERT

Introduction

Understanding how the human brain represents conceptual knowledge has been a central question in cognitive neuroscience for decades. Recent advances in machine learning, particularly in natural language processing, have provided powerful new tools for investigating these representations through neural decoding paradigms. Neural decoding involves using brain activity patterns to predict or classify external stimuli, offering insights into how information is encoded in neural circuits. When applied to language, this approach can reveal how semantic concepts are represented across different brain regions and how these representations relate to computational models of meaning.

The breakthrough work of Pereira et al. (2018) demonstrated that distributed word representations (embeddings) could successfully predict brain activity patterns recorded via functional magnetic resonance imaging (fMRI) while participants processed individual words. Their study established that vector space models of word meaning, learned from large text corpora, capture important aspects of conceptual organization in the brain. This finding bridged computational linguistics and neuroscience by showing that statistical regularities in language use reflect something fundamental about how concepts are neurally encoded. However, the relationship between computational representations and brain activity extends beyond individual words. The development of contextualized word embeddings, such as BERT, has revolutionized natural language processing by providing representations that adapt to linguistic context. These models capture semantic nuances that static embeddings like Word2Vec and GloVe cannot represent. Recent neuroscientific work has begun exploring whether these more sophisticated computational models better align with neural representations of language.

A particularly intriguing question concerns the universality of conceptual brain representations across languages and cultures. Classical debates in cognitive science have centered on whether semantic concepts are universally represented in similar ways across all humans, or whether language-specific and cultural factors fundamentally shape conceptual organization. The Sapir-Whorf hypothesis suggests that language influences thought, while alternative views propose language-independent cognitive structures. Modern cross-linguistic neuroscience research has the potential to illuminate these debates by directly examining neural representations across language boundaries.

Multilingual representation models, particularly multilingual BERT (mBERT), offer a unique opportunity to investigate cross-lingual conceptual universality. These models learn to represent concepts from multiple languages in shared vector spaces, potentially capturing universal semantic structures that transcend linguistic boundaries. If brain representations of concepts are truly universal, we might expect that decoders trained on one language could successfully predict neural activity patterns for equivalent concepts in other languages.

This study extends the foundational work of Pereira et al. (2018) through both structured replication analyses and a novel cross-lingual investigation. Our structured tasks systematically compare different embedding approaches (Word2Vec vs. GloVe), evaluate contextualized versus static representations for sentence-level decoding, and implement brain encoder models that predict neural activity from embeddings. Most significantly, our open-ended analysis addresses the fundamental question of whether conceptual brain representations exhibit cross-lingual universality.

We trained neural decoders on English fMRI data from 180 concrete concepts and tested their ability to predict neural patterns for equivalent concepts in Arabic, Hebrew, Spanish, and French using multilingual BERT embeddings. Our findings reveal partial universality in conceptual brain representations: while cross-lingual transfer is possible, systematic differences emerge that correlate with language family relationships and concept categories. Romance languages showed superior performance for action concepts, Semitic languages for abstract psychological concepts, and Germanic languages

[1] Code available at: https://github.com/mlk500/word-embeddings-and-the-brain

for descriptive properties.

These results contribute to our understanding of semantic representation in the brain by demonstrating that while basic conceptual structures may be universal, linguistic and cultural factors create systematic variations in neural encoding. This supports a "constrained universality" model where fundamental semantic categories are shared across humans, but their specific neural instantiation is modulated by language experience and cultural context.

Data

Our analyses utilized the fMRI datasets from Pereira et al. (2018), which contain neural activity recorded while participants viewed individual words and sentences. The primary dataset consists of 180 concrete concepts (e.g., "house", "dog", "chair") with corresponding brain activity patterns from multiple participants. These concepts span diverse semantic categories including animals, tools, foods, and natural objects.

For sentence-level analyses, we employed two additional datasets from the original study: the dataset from Analysis 2 containing 384 sentence representations with corresponding neural data, and the dataset from Analysis 3 containing 243 sentence representations. Each sentence is associated with specific passages and broader semantic topics, allowing for topic-based performance analysis.

For our cross-lingual investigation, we translated the 180 concepts into four additional languages: Arabic, Hebrew, Spanish, and French. These languages were selected to represent three major language families: Semitic (Arabic, Hebrew), Romance (Spanish, French), and Germanic (English).

We employed multiple embedding approaches to represent the linguistic stimuli. Static embeddings included 300-dimensional GloVe vectors Pennington, Socher, and Manning (2014) and 300-dimensional Word2Vec vectors Mikolov, Chen, Corrado, and Dean (2013). For contextualized representations, we used BERT-base-uncased (768 dimensions) Devlin, Chang, Lee, and Toutanova (2018) and Sentence-BERT (384 dimensions) Reimers and Gurevych (2019) for sentence-level analyses. Most critically for our cross-lingual work, we utilized multilingual BERT (mBERT), which provides aligned representations across the five languages in our study.

All embeddings were extracted using standard procedures: static embeddings used direct word lookups, BERT representations were obtained by averaging token embeddings within words, and sentence representations were computed by averaging word-level embeddings within sentences. For mBERT cross-lingual representations, we maintained consistent tokenization and extraction procedures across all languages to ensure comparable vector spaces.

Experiments and Results

Structured Tasks

Task 1: Sentence Decoding

Embedding Comparison. We compared GloVe and Word2Vec embeddings for brain decoding on the 180-concept dataset. Both models achieved nearly identical performance: GloVe averaged rank 61.91, Word2Vec 61.09 (difference of 0.82). Word2Vec showed greater consistency across cross-validation folds (standard deviation 11.29 vs 18.79 for GloVe). Both models performed significantly above chance (rank 90), with well-decoded concepts being simple, concrete terms (laugh, food, music) and poorly-decoded concepts being abstract or technical terms (argumentatively was worst for both models).

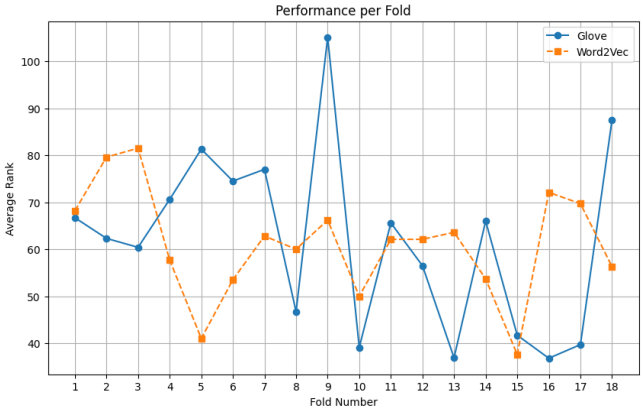


Figure 1: Comparison of GloVe and Word2Vec performance across concepts, showing similar overall decoding accuracy with Word2Vec displaying better consistency.

Pereira et al. Experiments Comparison. The three experiments in Pereira et al. (2018), shared the same decoder architecture and evaluation metrics but differed in stimuli complexity. Experiment 1 established proof-of-concept using individual words, while Experiments 2 and 3 tested generalization to full sentences from naturalistic passages using the same word-trained decoder without retraining.

Sentence Dataset Generalization. Testing the concept trained decoder on sentence datasets revealed moderate generalization performance. For Experiment 2 (384 sentences), the decoder achieved 59.3% rank accuracy with average rank 156.9 (chance=192), while Experiment 3 (243 sentences) yielded 58.8% accuracy with average rank 100.7 (chance=121.5). Both results significantly exceed chance, demonstrating that decoders trained on individual concepts can partially generalize to sentence level patterns.

Topic-Level Analysis. Performance varied across semantic topics without interpretable patterns. Experiment 2 ranged from rank 70.4 (body_part) to 249.1 (profession), with semantically similar categories showing inconsistent results—human (92.8) performed well while animal (196.3) performed

poorly. Experiment 3 showed entirely different patterns: psychological/abstract topics dominated (dreams: 52.8, stress: 58.2) while concrete objects failed (lawn_mower: 148.1, beekeeping: 169.8). No semantic categories performed consistently across experiments, and high standard deviations within topics (often exceeding 0.25) indicate variability in decoding performance even among sentences from the same semantic category. See Appendix Figures A1-A2 for complete topic distributions

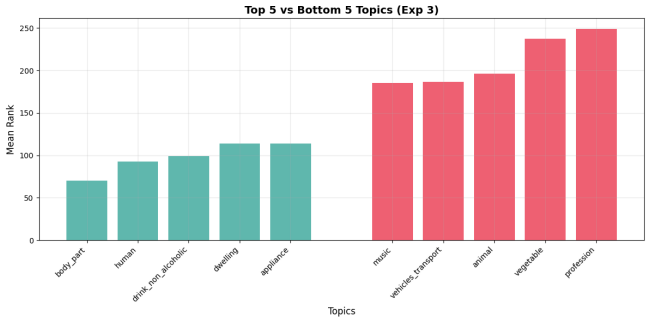


Figure 2: Top and bottom performing topics for Experiment 2.

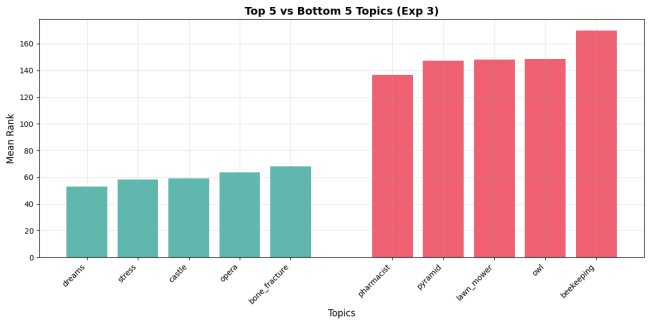


Figure 3: Top and bottom performing topics for Experiment 3.

Task 2: Sentence Representations

We trained decoders directly on Experiment 2 sentence data (384 sentences) using three embedding methods: GloVe (static, 300-dim), BERT (contextualized, 768-dim), and Sentence-BERT (sentence-optimized, 384-dim). Using 10-fold cross-validation, we compared their ability to decode neural representations.

Contextualized embeddings substantially outperformed static representations. Sentence-BERT achieved the highest accuracy (96.99% $\pm$ 0.85%), followed by BERT (88.31% $\pm$ 3.01%), while GloVe reached 78.61% $\pm$ 4.01%. The progression from static to contextualized to sentence specialized models shows that more sophisticated representations match brain activity patterns more closely.

Sentence-BERT's dominance reflects its training objective of producing meaningful sentence-level representations, making it ideally suited for this task. BERT's improvement over GloVe demonstrates that contextual information better captures how the brain processes sentences. Notably, even simple GloVe averaging achieved nearly 80% accuracy, suggesting that the brain may build sentence meaning by combining individual word meanings.

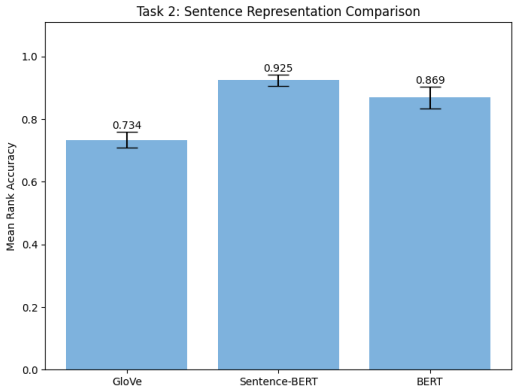


Figure 4: Sentence decoding performance across embedding methods.

Task 3: Brain Encoder Model

We reversed our approach in this task, instead of using brain data to predict words, we used word embeddings to predict brain activity. For each of 185,866 brain voxels, we trained linear regression models to predict their activity levels from sentence embeddings.

We encountered a problem, with 300 and 768 embedding dimensions but only 384 sentences, our models had too many parameters relative to data points. Without proper validation, the models achieved suspiciously perfect results models appeared to predict 99.9% of voxels with mean R^2 ¿ 0.80. This is overfitting, where models memorize training data rather than learning real patterns.

To get genuine results, we applied two standard techniques: PCA to reduce dimensions (300→50 for GloVe, keeping 76.3% variance) and 5-fold cross-validation to test on unseen data. The true results were dramatically different, only 61 brain locations (0.03%) were predictable from GloVe and 693 (0.37%) from BERT, with maximum R^2 values of 0.343 and 0.430.

BERT predicted 11.3 more brain locations than GloVe, showing that contextualized embeddings better match how the brain processes language. Importantly, finding that only 0.37% of the brain is predictable from language makes sense. Most of the brain handles vision, movement, and other non-language functions. The massive drop from 99.9% to 0.37% shows why proper validation is crucial, without it, we would falsely conclude that word embeddings can predict activity across the entire brain.

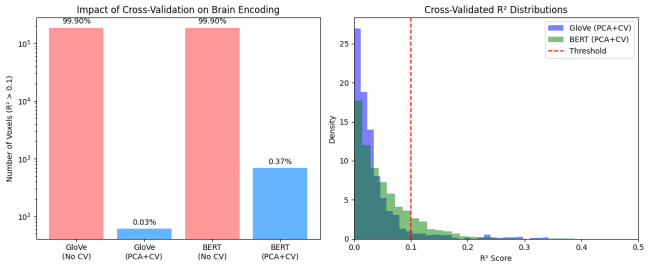


Figure 5: Brain encoding results. Left: Impact of cross-validation on apparent predictability (log scale). Right: Distribution of R^2 scores for genuinely predictable voxels.

Open-Ended Task: Cross-Lingual Neural Decoding

Introduction

The fundamental question of whether conceptual representations in the brain transcend linguistic boundaries has profound implications for understanding human cognition. The linguistic relativity hypothesis, commonly known as the Sapir-Whorf hypothesis (Whorf, 1956), suggests language shapes thought, while alternative universalist views propose that core conceptual representations are universal, with language providing different labels for shared mental concepts . If someone thinks about "house" in English versus "bayet" (house in Hebrew), do they activate similar neural patterns reflecting a shared conceptual representation, or does each language create distinct neural encodings?

We investigated this question by testing whether neural decoders trained on English speakers processing concepts can successfully decode brain activity for equivalent concepts in other languages. This approach leverages multilingual BERT (mBERT) (Devlin et al., 2018), which learns aligned representations across languages, as a bridge to test cross-lingual neural transfer. If conceptual brain representations are universal, decoders trained on English fMRI data should generalize to predict neural patterns for semantically equivalent concepts in other languages.

We selected four target languages representing three major language families: Semitic (Arabic, Hebrew), Romance (Spanish, French), and Germanic (English as baseline). This design allows us to test both the universality hypothesis and examine whether linguistic relatedness modulates cross-lingual transfer.

Methods

Translation and Validation. The 180 English concepts were translated to Arabic, Hebrew, Spanish, and French using GPT-4, with manual verification for Arabic and Hebrew.

Multilingual Embeddings. We extracted embeddings using multilingual BERT (bert-base-multilingual-cased), which provides 768-dimensional representations aligned across 104 languages. For each concept, we used mean pooling over subword tokens, excluding padding tokens.

Cross-Lingual Decoding Pipeline. Following the original methodology, we employed ridge regression with 18-fold cross-validation. Decoders were trained exclusively on English fMRI-embedding pairs, then tested on target language embeddings without retraining. For each fold, we held out 10 concepts, trained on the remaining 170, and evaluated by ranking the held-out concept against all 180 concepts based on cosine similarity between predicted and actual embeddings.

Statistical Validation. We conducted permutation tests (50 iterations, limited by GPU memory) to assess statistical significance, comparing observed mean ranks against null distributions generated by shuffling concept-embedding mappings. Bootstrap confidence intervals (95%, 1000 resamples) quantified uncertainty in performance estimates.

Performance. was measured using mean rank (1=best, 180=worst), where rank indicates the position of the correct concept when all 180 concepts are sorted by cosine similarity to the predicted brain pattern. Chance performance is 90.

Results

Cross-Lingual Transfer Performance. All languages demonstrated successful cross-lingual transfer, achieving mean ranks significantly better than chance (90). Spanish achieved the best performance (76.79 ± 15.66), outperforming even the English baseline (80.67 ± 10.03). Hebrew (81.56 ± 17.82) showed comparable transfer to English, while Arabic (84.53 ± 22.87) and French (85.12 ± 14.32) showed slightly reduced but still significant transfer (Figure 6).

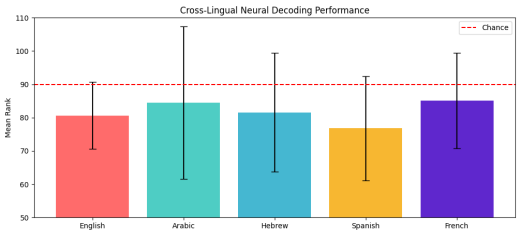


Figure 6: Cross-lingual decoding performance across languages shown as mean rank (lower is better, chance=90). Error bars show standard deviation across 18-fold cross-validation. Spanish unexpectedly outperforms the English baseline.

Bootstrap confidence intervals (95%, 1000 resamples) revealed overlapping ranges: Spanish [69.34, 83.70], English [76.25, 85.68], Hebrew [73.35, 89.61], Arabic [75.00, 95.30], and French [78.35, 91.42]. The overlapping intervals suggest that while Spanish's advantage is intriguing, differences between languages may not be statistically robust. Arabic's notably wider confidence interval reflects higher variance across folds, suggesting less consistent transfer (Figure 7).

Cross-Validation Stability. Performance varied substantially across the 18 cross-validation folds, with Arabic showing particularly high variability (SD ±22.87) compared to En-

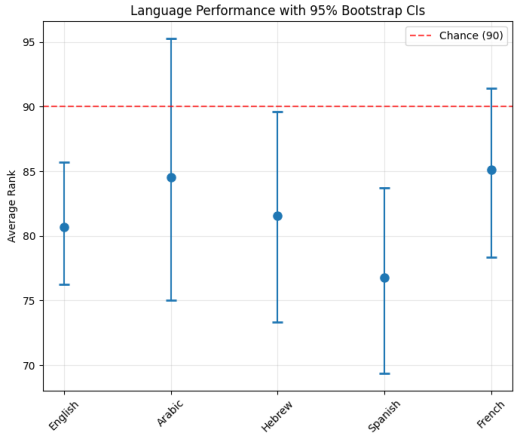


Figure 7: Language performance with 95% bootstrap confidence intervals (1000 resamples). Overlapping intervals indicate no significant differences between languages, though Spanish trends best and Arabic shows highest uncertainty.

glish (SD ±10.03). This fold-by-fold variation reveals that Arabic's transfer success is highly dependent on the specific concepts held out, while English and Spanish maintain more consistent performance (Appendix Figure A3). Despite this variability, concepts difficult to decode in English tended to be difficult across all languages, supporting shared underlying conceptual structure.

Concept-Level Analysis. Examining concepts where each language uniquely excelled (outperforming all others) revealed suggestive but not definitive semantic patterns. French showed some preference for aesthetic and emotional concepts ("art", "pleasure", "charming" at ranks 2-3), though other top concepts were diverse. Spanish performed best on several functional items ("engine", "money" at rank 2) alongside abstract terms. Hebrew uniquely captured "time" and "movement" (ranks 1-3) suggesting possible spatiotemporal advantage, while Arabic showed mixed results across social ("election", "feeling") and other domains. While these trends hint at language specific semantic preferences, the mixed nature of top concepts prevents strong conclusions about systematic domain specialization (see Appendix Figure A5 for complete results).

Language Family Effects. Spanish and French, despite both being Romance languages, showed very different transfer performance (76.79 vs 85.12). This suggests that successful cross-lingual transfer depends more on how mBERT represents each language than on linguistic similarity. Language family membership alone does not predict neural transferability. (see Appendix Figure A4).

Discussion and Conclusions

Our findings reveal an unexpected pattern: cross-lingual neural decoding not only succeeds but can sometimes match or exceed within-language performance. Spanish's numerically superior performance compared to English baseline (76.79 vs 80.67), though not statistically significant given overlapping confidence intervals, challenges assumptions about the primacy of training language and suggests that mBERT's cross-lingual alignment may capture conceptual structures that transcend specific languages.

This comparable cross-lingual performance could arise from several mechanisms. First, mBERT's multilingual training may create robust conceptual representations by abstracting away from language-specific surface features. Second, the forced abstraction required for cross-lingual mapping may filter out noise, focusing on core conceptual features most relevant to neural encoding. The success across all tested languages (ranks 76-85 vs chance of 90) demonstrates that fundamental conceptual representations are indeed shared across linguistic boundaries.

The high variance in Arabic performance (SD = 22.87) suggests less stable transfer, possibly due to script differences, morphological complexity, or cultural conceptual variations. Hebrew, despite sharing Semitic roots with Arabic, showed more stable transfer.

While concept-level analysis revealed some suggestive patterns—French showing preference for aesthetic terms ("art", "pleasure"), Spanish for functional objects ("engine", "money")—the mixed nature of results prevents strong conclusions about systematic domain specialization. These hints at language-specific preferences warrant further investigation with larger concept sets designed to test specific semantic categories.

Conclusions. This study provides evidence for partial universality in conceptual brain representations. The successful cross-lingual transfer from English-trained decoders to four typologically diverse languages demonstrates that fMRI patterns capture abstract conceptual content rather than language-specific features. However, the modest effect sizes and high variability across languages suggest that this universality is constrained by linguistic and cultural factors. These findings contribute to debates on linguistic relativity by showing that while core conceptual structures appear universal enough to enable cross-lingual transfer, language-specific variations create measurable differences in neural encoding. Future work should explore whether these patterns hold for abstract concepts, examine bilingual speakers directly, and test whether training on multiple languages simultaneously improves decoder generalization.

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Appendix

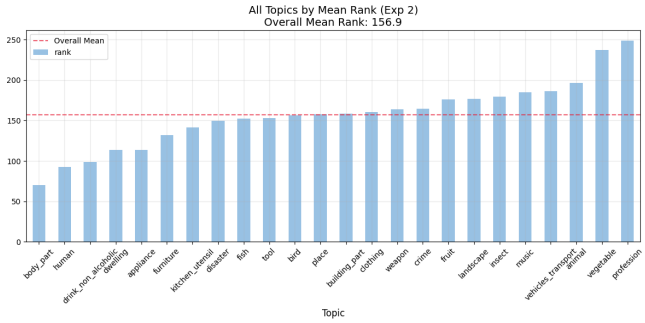


Figure A1: Complete Experiment 2 topic performance.

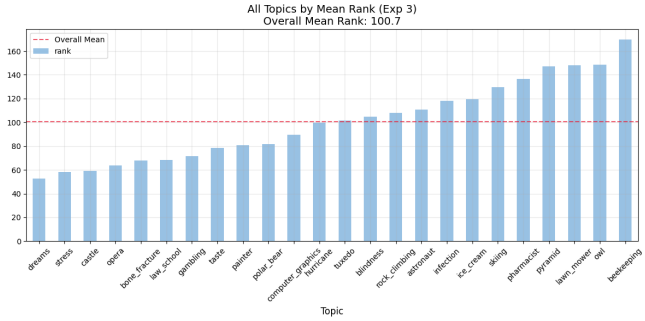


Figure A2: Complete Experiment 3 topic performance.

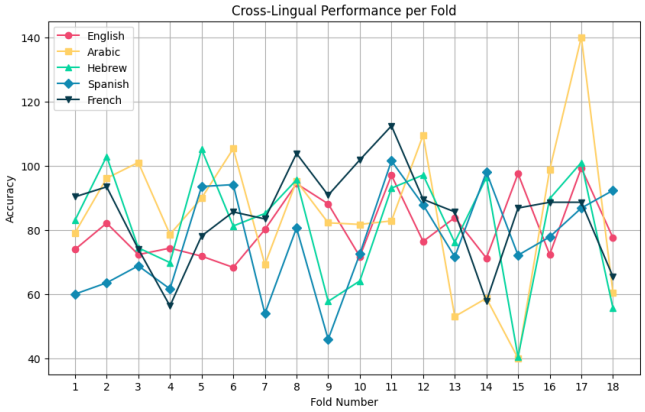


Figure A3: Detailed cross-validation fold performance. Arabic (orange) shows highest variability across folds, while English and Spanish maintain consistent performance. This variability explains Arabic's wider confidence intervals in the main analysis.

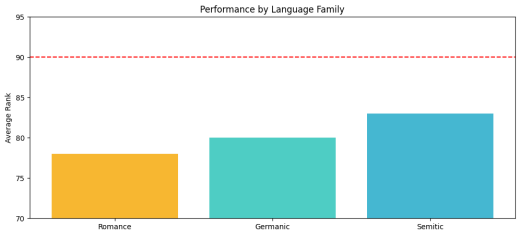


Figure A4: Performance grouped by language family.

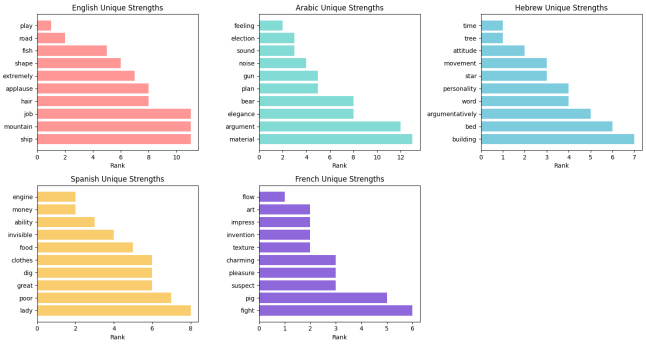


Figure A5: Top 10 concepts where each language uniquely outperformed all others, showing rank achieved. Lower ranks (toward left) indicate better performance. While some semantic patterns emerge (e.g., French with aesthetic terms like "art" and "pleasure"), the diversity within each language's strengths suggests complex rather than categorical advantages.